# Stock Researcher Agent

INTEGRATED OPERATIONS MANUAL & EXECUTION RUNBOOK

PRODUCTION LIVE | REVISION V3.0 UNIFIED

Updated: 11 October 2026 • Region: SG / IN

Repo: `appurajacool2015/stock-researcher-agent`

> **Core Invariant:** The Render API is the deterministic financial calculation and data-collection engine. The LLM (Requesty / OpenClaw) must *never* calculate, estimate, or fabricate financial indicators. All numerical data, metrics, and quality flags originate solely from the canonical `ResearchResult v1` payload.

## 1. IMMEDIATE UP-TO-SPEED & SMOKE TEST PROCEDURES

Follow this sequential command sequence to verify the complete operational stack after inactivity or deployment updates:

```
# Step 1: Connect to Oracle Cloud VM & inspect OpenClaw gateway status ssh -i ssh-key-2026-08-27.key ubuntu@129.225.120.74 systemctl --user status openclaw-gateway --no-pager # Step 2: Validate Requesty LLM router connectivity (run on the VM) openclaw agent --agent main --message "Reply with exactly: Requesty connection OK." --timeout 60 # Step 3: Verify Render API live health probe curl -fsS https://stock-researcher-agent.onrender.com/health # Step 4: Validate live market data pipeline using the single-line jq extractor curl -fsS 'https://stock-researcher-agent.onrender.com/research/INFY' | jq '{symbol: .symbol, status: .overall_status, cmp: .technicals.cmp.value, quality: .data_quality_summary}'
```

## 2. MASTER ACCOUNT & RESOURCE DIRECTORY

| Service / Role | Account Identifier / Endpoint | Operational Notes & Guardrails |
|---|---|---|
| **Oracle Cloud VM** | Account: `ajjw2025` (Root)<br>IP: `129.225.120.74` | Host for OpenClaw Gateway. Access via Linux user `ubuntu` using SSH key. Never expose private keys. |
| **Render Production API** | `ajjw2025@gmail.com`<br>`stock-researcher-agent.onrender.com` | FastAPI Uvicorn host (Singapore). Free tier spins down after ~15m idle; warm-up takes 40–55s. |
| **Requesty LLM Router** | `ajjw2025@gmail.com`<br>URL: `https://router.requesty.ai/v1` | Active Model: `google/gemma-4-31b-it`. Key stored in `~/.openclaw/.env` as `REQUESTY_API_KEY`. |
| **GitHub Source Repo** | `appurajacool2015@gmail.com`<br>`appurajacool2015/stock-researcher-agent` | Single source of truth for code. Auto-deploys to Render on push to `main` branch. |
| **Setup Discussions** | ChatGPT & Gemini Workspaces | Shared design contexts: ChatGPT Thread • Gemini Thread |

## 3. QUICK OPERATIONAL TRIAGE & PITFALLS

**NOTICE** **Status PARTIAL is Valid**

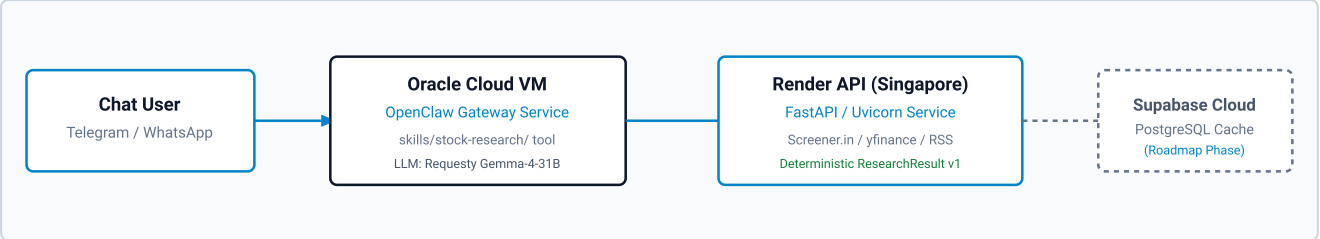
For zero-debt companies (e.g., TCS), metrics like debt-to-equity and promoter pledge are naturally non-existent. These mark as `UNAVAILABLE`, yielding an overall `PARTIAL` status. This is expected behavior, not a pipeline bug.

**ALERT** **Render Free Tier Cold Starts**

Initial requests after 15 min idle take 40–55 seconds. Configure OpenClaw tool timeouts to **75 seconds**. Do not use `GET /` for uptime monitoring (returns 404); use `GET /health`.

## 4. GLOBAL RUNTIME TOPOLOGY & COMMUNICATION ARCHITECTURE

The agent enforces a strict boundary between user conversation orchestration and deterministic data acquisition:




## 5. RENDER PRODUCTION SPECIFICATIONS & BUILD CONFIGURATION

| Configuration Key | Configured Value | Operational Description |
|---|---|---|
| **Repository & Branch** | `appurajacool2015/stock-researcher-agent` (`main`) | Auto-syncs and builds upon git push. Commit `7ef7c63` validated live. |
| **Runtime & Region** | Python 3 • Singapore (Southeast Asia) | Optimal latency profile for NSE / BSE market scraping adapters. |
| **Build Command** | `pip install -r requirements.txt` | Installs FastAPI, Uvicorn, yfinance, BeautifulSoup4, requests. |
| **Start Command** | `uvicorn interfaces.api:app --host 0.0.0.0 --port $PORT` | Binds to Render's dynamic container port (observed port: 10000). |
| **Health Check Route** | `/health` | Prevents container recycle alerts caused by default root HEAD 404s. |

## 6. OPENCLAW SKILL WIRING (ORACLE CLOUD VM)

Configure `skills/stock-research/SKILL.md` on the VM to invoke Render directly via HTTPS rather than relying on local subprocesses:

```
    # skills/stock-research/SKILL.md execution binding import requests def
execute_stock_research(symbol: str) -> dict: # 1. Sanitize & normalize ticker input clean_symbol =
symbol.strip().upper().replace("$", "") url = f"https://stock-researcher-
agent.onrender.com/research/{clean_symbol}" # 2. Invoke Render microservice with cold-start
allowance (75s) response = requests.get(url, timeout=75) response.raise_for_status() # 3. Return
verified deterministic ResearchResult v1 payload return response.json()
```

## 7. COMPREHENSIVE TROUBLESHOOTING & RECOVERY MATRIX

| Observed Symptom | Probable Root Cause | Standard Recovery & Fix Procedure |
|---|---|---|
| **Gateway Timeout (>50s)** | Render Free tier container spinning up after inactivity. | Set OpenClaw tool timeout to `75s`. Issue a warm-up GET request to `/health` before research tasks. |
| **HTTP 404 on Root Route** | Probing `/` or `HEAD /` instead of defined route. | Update probe path to `/health`. FastAPI routes exist at `/health`, `/research/{sym}`, and `/docs`. |
| **Invalid Character / jq Error** | Line-continuation backslashes copied into shell. | Use single-line quote syntax: `curl -fsS 'URL' \| jq '{symbol: .symbol, ...}'`. |
| **LLM Output Missing Data** | Agent attempting to synthesize data without payload. | Ensure `SKILL.md` is properly wired to Render API. Check gateway logs via `journalctl --user -u openclaw-gateway`. |
| **Requesty Auth Failure** | Missing or expired API key in OpenClaw environment. | Verify REQUESTY_API_KEY exists in `~/.openclaw/.env`. Legacy AWS Bedrock is deprecated; do not re-enable. |

## 8. ROADMAP: SUPABASE POSTGRESQL PERSISTENCE ARCHITECTURE

To prevent redundant external scraping, provide a 6-hour caching layer, and preserve longitudinal historical research across user chat sessions, execute the following SQL schema in Supabase:

```
-- Persistence Migration Schema: stock_research_runs CREATE TABLE IF NOT EXISTS stock_research_runs ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), symbol VARCHAR(20) NOT NULL, exchange VARCHAR(10) DEFAULT 'NSE', company_name TEXT, overall_status VARCHAR(20) NOT NULL, data_quality_summary JSONB NOT NULL, result_payload JSONB NOT NULL, created_at TIMESTAMPTZ DEFAULT NOW() ); -- Indexes for fast 6-hour lookup and auditing CREATE INDEX idx_stock_runs_symbol_created ON stock_research_runs (symbol, created_at DESC); CREATE INDEX idx_stock_runs_status ON stock_research_runs (overall_status);
```

## 9. DEVELOPER REFERENCE & LOCAL CHEATSHEET

| Development Task | Command / Location | Expected Output / Action |
|---|---|---|
| **Execute Unit Tests** | `venv-market/bin/python -m unittest tests/test_api.py` | 12 tests OK (Deterministic payload validations pass). |
| **Run Local API Server** | `venv-market/bin/python -m interfaces.api` | FastAPI listening on `http://0.0.0.0:8000`. |
| **Deploy to Production** | `git pull --rebase origin main && git push origin main` | Triggers Render webhook build and automated deployment. |
| **Interactive Docs UI** | `https://stock-researcher-agent.onrender.com/docs` | Swagger UI for interactive REST endpoint execution. |